FINAL PROJECT DOCUMENT – CARMARK CUSTOMER ANALYTICS DATABASE

CarMark: Customer Analytics Database

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ISM5560 Final Project

Introduction:

CarMark is a long-standing community-oriented automotive dealership known for creating highly personalized relationships with its customers. Historically, sales representatives maintained handwritten notes, spreadsheets, and paper files documenting customer information such as lifestyle traits, financial circumstances, prior vehicle ownership, and preferences identified during consultations. While this informal method fosters strong customer relationships, it does not scale, lacks data consistency, limits analytical insight, and cannot support predictive modeling or targeted marketing campaigns.

This project modernizes CarMark's customer data management by designing and implementing a Snowflake-based relational database that centralizes customer information, supports historical tracking, enables analytical computations, and integrates with automated business processes. The goal is to equip CarMark with a scalable, analytics-ready data foundation that enhances customer understanding, marketing precision, and sales outcomes.

The system incorporates effective-dated (SCD2) history for key attributes, a financial readiness model, customer segmentation supported by propensity scores, and Snowflake-native features such as MERGE-based SCD logic, VARIANT data, window functions, stored procedures, and data quality checks. Together, these components support a complete customer lifecycle, from lead creation to conversion to long-term retention.

Business Case and Objectives:

CarMark faces several data challenges that limit performance and inhibit growth:

- Customer information is inconsistent or incomplete across spreadsheets and handwritten notes.
- No unified system exists for tracking customer intent or monitoring behavioral changes over time.
- Sales teams cannot quantify which customers are most likely to purchase or which segments they best align with.

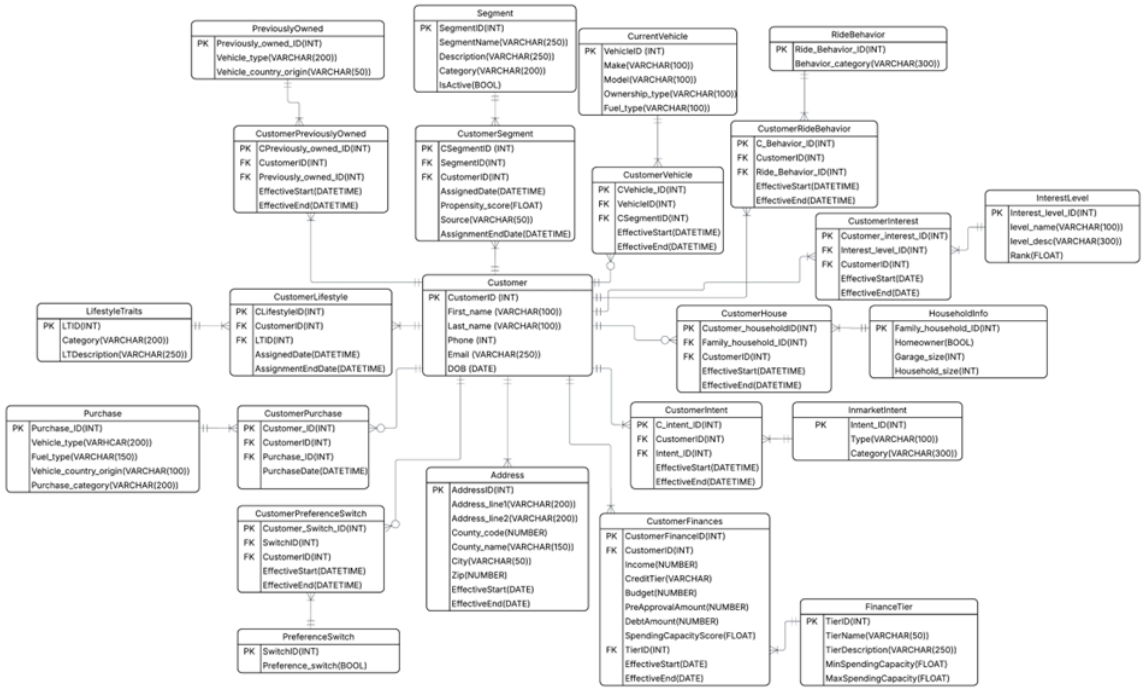
- Leadership lacks visibility into customer trends, loyalty patterns, market preferences, and financial readiness.

To address these gaps, our design goals were to:

- Consolidate all customer attributes into a clean relational schema.
- Support many-to-many relationships across lifestyle traits, behaviors, intents, and prior ownership.
- Track changes over time using effective-dated SCD2 structures.
- Implement a financial tier model derived from raw financial inputs.
- Generate segment-level propensity scores that support targeted outreach.
- Model a realistic lifecycle of leads, conversions, and re-engagement.
- Leverage Snowflake capabilities such as MERGE, VARIANT, FLATTEN, CTEs, window functions, and stored procedures.
- Provide meaningful business insights through analytical SQL queries.

Final Database Schema:

The final ERD contains 25 entities that capture the full scope of customer information, including demographics, addresses, lifestyle traits, ride behaviors, interests, financial information, vehicle ownership, purchase history, and segmentation. Associative bridge tables support many-to-many relationships, and multiple tables use effective dating to ensure that CarMark never loses historical context.



Key design principles include:

- All attributes that can change over time are stored in SCD2-enabled tables (CustomerSegment, CustomerFinances, CustomerIntent, CustomerPreviouslyOwned, CustomerVehicle, CustomerHouse).
- Current and historical versions coexist to support as-of analysis.
- Segmentation is handled through a flexible CustomerSegment bridge that stores multiple segments per customer with propensity scores.
- The schema is normalized for analytics and future scalability.

Lead Lifecycle Flow:

We developed a complete lead lifecycle to represent how customer information enters, evolves, and informs decisions within CarMark's operations.

Lead Generation

A lead is created when a customer fills out an interest form, visits the dealership, or interacts with the CRM system. The database captures:

- Demographics (Customer)
- Address and household details (CustomerAddress, CustomerHouse)
- Financial inputs (CustomerFinances)
- In-market intent (CustomerIntent)
- Lifestyle and behavior indicators (CustomerLifestyle, CustomerRideBehavior)
- Prior ownership (CustomerPreviouslyOwned)

Lead Conversion

A conversion occurs when a customer decides to make a purchase or expresses strong buying intent. This event triggers:

- A new CustomerPurchase record
- Updates to CustomerVehicle for the purchased vehicle
- PreferenceSwitch records if preferences shift
- Updates to CustomerSegment when segment alignment changes
- Recalculation of financial tier through the stored procedure

Customer Evolution Post-Conversion

Returning customers progress through new cycles of intent, preference changes, financial adjustments, and future purchases. Effective-dated tables preserve all historical versions, enabling CarMark to track changes in:

- Segment alignment
- Vehicle preferences
- Financial capacity
- Lifestyle traits
- Intent signals

This supports predictive analytics such as identifying upgrade opportunities or forecasting repeat purchases.

Snowflake Architecture and Implementation:

The Snowflake implementation included the creation of tables, constraints, surrogate keys, sequences, and SCD2 logic. The design also incorporates several Snowflake-native components:

- MERGE-based SCD2 updates for CustomerSegment
- MERGE-based upserts for CustomerFinances
- Stored procedure to recalculate financial tiers
- VARIANT and FLATTEN for semi-structured preference data
- Window functions for ranking and analytic insights
- CTEs for summarization
- Data quality checks to validate referential integrity
- Surrogate keys to manage versioned records

SCD2 Implementation for CustomerSegment:

A staging table was created to load new propensity scores. A MERGE operation updates the historical record by closing it and inserts a new current record with updated values. This versioning structure maintains complete history over time.

Customer 1 and Customer 5 were used to demonstrate improved propensity scores in the Luxury and Adventure segments. The SCD2 history query displayed both historical and current versions.

SQL screensho of SCD2:

```sql
-- Show SCD2 history for updated customers
SELECT
    CustomerID, SegmentID, Propensity_score,
    AssignedDate, AssignmentEndDate,
    CASE WHEN AssignmentEndDate IS NULL THEN 'Current' ELSE 'Historical' END AS Status
FROM CustomerSegment
WHERE (CustomerID = 1 AND SegmentID = 3)
   OR (CustomerID = 5 AND SegmentID = 2)
ORDER BY CustomerID, SegmentID, AssignedDate;
```

Results (just now)

Table | Chart — 4 rows — 155ms

	CUSTOMERID	SEGMENTID	PROPENSITY_SCORE	ASSIGNEDDATE	ASSIGNMENTENDDATE	STATUS
1	1	3	0.88	2023-01-01 00:00:00.000	2025-12-02 15:59:11.885	Historical
2	1	3	0.93	2025-12-02 16:07:07.172	null	Current
3	5	2	0.81	2023-01-01 00:00:00.000	2025-12-02 15:59:11.885	Historical
4	5	2	0.86	2025-12-02 16:07:07.172	null	Current

Analytical Model for Financial Tier:

An analytical model computes each customer's Spending Capacity Score (SCS) using:

- Income
- Budget
- Pre-approval amount
- Debt amount
- Credit-tier offsets

The score maps into five financial tiers: Premium (1), Strong (2), Moderate (3), Limited (4), and Restricted (5). A stored procedure recalculates these tiers when financial data changes.

SQL screenshot of recalculated financial data:

Results (1 minute ago)

Table | Chart — 25 rows — 120ms

	CUSTOMERID (1 – 25)	SPENDINGCAPACITYSCORE (18 – 96.5)	TIERID (1 – 5)
1	3	96.5	1
2	2	77	2
3	11	73	2
4	8	62	3
5	18	61.1	3
6	13	59.3	3
7	22	59.2	3
8	1	57.4	3
9	21	56.9	3
10	4	46	3

MERGE-Based UPSERT for CustomerFinances:

A MERGE statement ensures that the most recent financial information replaces outdated values, maintaining accuracy across segmentation and tier calculations.

SQL screenshot: Financial MERGE upsert result.

```
        );
SELECT
    CustomerFinanceID,
    CustomerID,
    Income,
    Budget,
    PreApprovalAmount,
    DebtAmount,
    EffectiveStart,
    EffectiveEnd
FROM CustomerFinances
WHERE CustomerID IN (1, 5)
ORDER BY CustomerID, EffectiveStart;
```

Results (just now)

Table | Chart — 2 rows — 430ms

	CUSTOMERFINANCEID	CUSTOMERID	INCOME	BUDGET	PREAPPROVALAMOUNT	DEBTAMOUNT	EFFECTIVESTART	EFFECTIVEEND
1	1	1	98000	950	30000	15000	2023-01-01	null
2	5	5	76000	700	24000	20000	2020-04-01	null

Semi-Structured Data: VARIANT + FLATTEN:

To support CRM-style enrichment data, a Preferences VARIANT column was added to the Customer table. JSON values store preferred brands and budget ranges, and FLATTEN extracts nested elements for analysis.

Example JSON:

- "preferred_brands": ["Honda", "BMW"]
- "budget_range": {"min": 20000, "max": 45000}\

```
-- Flatten query against stored JSON variant
SELECT
    CustomerID,
    brand.value::STRING AS PreferredBrand
FROM Customer,
     LATERAL FLATTEN(input => Preferences:preferred_brands) brand;
```

Results (3 minutes ago)

Table | Chart — 2 rows — 378ms

	CUSTOMERID	PREFERREDBRAND
1	1	Honda
2	1	BMW

Data Quality Checks:

The system includes multiple integrity checks:

- Null foreign key detection in CustomerSegment
- Invalid TierID detection in CustomerFinances
- Negative financial value checks
- Overlapping SCD2 ranges using window functions

All checks returned zero violations, confirming proper implementation.

```sql
-- Check for NULL foreign keys
SELECT *
FROM CustomerSegment
WHERE CustomerID IS NULL OR SegmentID IS NULL;

-- Check for invalid TierIDs
SELECT *
FROM CustomerFinances
WHERE TierID NOT BETWEEN 1 AND 5;

-- Check for impossible financial values
SELECT *
FROM CustomerFinances
WHERE Income < 0 OR Budget < 0 OR PreApprovalAmount < 0;
-- produced no results
```

Results (just now)

Table | Chart | 0 rows

CUSTOMERFINANCEID	CUSTOMERID	INCOME	CREDITTIER	BUDGET	PREAPPROVALAMOUNT	DEBTAMOU

Query produced no results

Decision-Relevant Business Queries:

We developed analytical queries to support strategic decision-making. These include:

- Identifying high-propensity, high-tier luxury prospects
- Counting customers by financial tier
- Calculating average propensity scores by segment
- Listing customers with strong upgrade intent
- Ranking loyalty based on historical vehicle ownership

```sql
-- 1. Top luxury prospects (segment + finance)
SELECT
    cs.CustomerID,
    cs.Propensity_score AS LuxuryScore,
    cf.TierID
FROM CustomerSegment cs
JOIN CustomerFinances cf ON cs.CustomerID = cf.CustomerID
WHERE cs.SegmentID = 3
  AND cs.AssignmentEndDate IS NULL
ORDER BY LuxuryScore DESC, cf.TierID;

-- 2. Customer distribution by financial tier
SELECT TierID, COUNT(*) AS CustomerCount
```

Results (just now)

Table | Chart — 25 rows — 867ms

	CUSTOMERID (1–25)	LUXURYSCORE (0.21–0.94)	TIERID (1–5)
1	21	0.94	3
2	1	0.93	3
3	13	0.93	3
4	8	0.92	3
5	3	0.91	1
6	18	0.9	3
7	24	0.89	4
8	12	0.89	4

Multi-Factor Prospect Scoring:

A multi-factor query combines:

- Segment propensity
- Spending Capacity Score
- Count of active in-market intent signals

This provides a unified view of high-value prospects and supports targeted marketing and sales prioritization.

```sql
SELECT
    c.CustomerID,
    c.First_Name,
    c.Last_Name,
    cs.Propensity_score AS LuxuryFit,
    cf.SpendingCapacityScore,
    (
        SELECT COUNT(*)
        FROM CustomerIntent ci
        WHERE ci.CustomerID = c.CustomerID
          AND ci.EffectiveEnd IS NULL
    ) AS ActiveIntentSignals
FROM Customer c
JOIN CustomerSegment cs ON c.CustomerID = cs.CustomerID
    AND cs.SegmentID = 3
    AND cs.AssignmentEndDate IS NULL
JOIN CustomerFinances cf ON c.CustomerID = cf.CustomerID
ORDER BY LuxuryFit DESC, SpendingCapacityScore DESC;
```

Results (just now)

Table | Chart — 25 rows — 604ms

	CUSTOMERID (1–25)	FIRST_NAME (Alex 4.0%, Ava 4.0%, +23 more)	LAST_NAME (Adams 4.0%, Baker 4.0%, +23 more)	LUXURYFIT (0.21–0.94)	SPENDINGCAPACITYSCORE (18–96.5)	ACTIVEINTENTSIGNALS (1–1)
1	21	Jacob	Carter	0.94	56.9	1
2	13	Henry	Young	0.93	59.3	1
3	1	Alex	Johnson	0.93	57.4	1
4	8	Sophia	Martinez	0.92	62	1
5	3	James	Nguyen	0.91	96.5	1
6	18	Ava	Perez	0.9	61.1	1

SCD2 Customer History and As-of Analysis:

The database supports “as-of” queries to reconstruct a customer’s state at a particular point in time. Customer 1’s Luxury segment history was used as an example to demonstrate how segment alignment evolves.

As-of SCD2 history output:

```
*I to generate
SELECT
    CSegmentID, CustomerID, SegmentID,
    Propensity_score, AssignedDate, AssignmentEndDate
FROM CustomerSegment
WHERE CustomerID = 1
  AND SegmentID = 3
ORDER BY AssignedDate;
```

Results (just now)

Table | Chart — 2 rows — 159ms

	CSEGMENTID	CUSTOMERID	SEGMENTID	PROPENSITY_SCORE	ASSIGNEDDATE	ASSIGNMENTENDDATE
1	3	1	3	0.88	2023-01-01 00:00:00.000	2025-12-02 15:59:11.885
2	1101	1	3	0.93	2025-12-02 16:07:07.172	null

Conclusion:

The CarMark Customer Analytics Database transforms the dealership’s legacy information structure into a modern, scalable, analytics-enabled platform. Through Snowflake’s capabilities and a relational schema designed for both transactional and analytical use, CarMark can now:

- Track changes in customers over time
- Apply data-driven segmentation using propensity scores
- Automate financial readiness calculations
- Integrate structured and semi-structured customer data
- Support full lifecycle analytics, from lead creation to conversion to repeat business

The system meets all project objectives and provides a strong foundation for predictive modeling and personalized marketing.